Norwegian University of Science
and Technology
Department of Electronic Systems

TFE4130 Wave Propagation

Exercise set 1 & 2

1 Locomotive and the Doppler effect

a) The velocity of a locomotive is 100 km/t along a straight railway. If its whistle has frequency $f = 440$ Hz, what is the frequency as heard by a stationary observer close to the railway? Assume that the locomotive approaches the observer (and that the distance between the railway and the observer is much less than the distance between the locomotive and the observer).

The speed of sound in air is taken to be 340m/s.

b) The Doppler radar used by police to check the speed of a car, emits an electromagnetic wave of frequency f. We assume that the road is straight, and that the radar is located close to the road. After the radar signal is reflected by an approaching car, what will be the detected frequency as seen by the radar? You may certainly assume that the velocity is much less than the speed of light, to simplify the expression.

Hint: Since the car reflects the radar wave, you may imagine that it first detects the wave (and thereby observes the frequency), and then emits a wave with this frequency. The relativistic Doppler frequency shift for electromagnetic waves is

$$f' = f\sqrt{\frac{1-u/c}{1+u/c}}. \tag{1}$$

2 Plane sound waves

We have mounted a loudspeaker element, of diameter 4 inches = 101.6 mm, in the end of a circular tube (with the same diameter as the loudspeaker), see Figure 1.

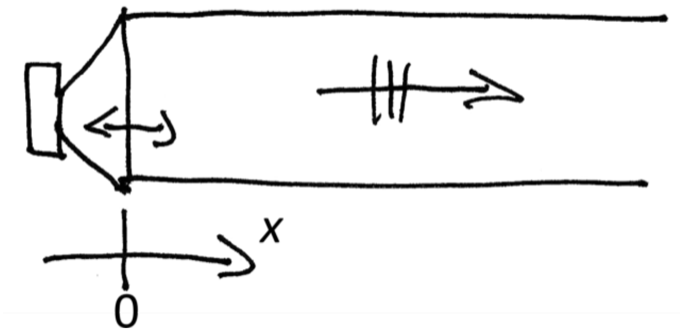

Figure 1: Speaker at the end of a circular tube.

The loudspeaker element is vibrating sinusoidally with the frequency 100 Hz, that is, the loudspeaker surface displacement is

$$\zeta_{\text{lsp}}(t) = \hat{\zeta}\sin(\omega t), \tag{2}$$

with $\hat{\zeta} = 0.5\,\text{mm}$.

a) Write the displacement on the complex-exponential form instead, such that

$$\zeta_{\text{lsp}}(t) = \text{Re}|\zeta_{\text{lsp,complex}}(t)|, \tag{3}$$

where $\zeta_{\text{lsp,complex}}(t) = Ae^{j\omega t}$.
What is A, knowing that A might be a complex number?

b) What is the vibration velocity, $v_{\text{lsp}}(t)$? Derive it from the displacement functions, both on the real form and the complex-exponential form.
Hint:

$$v_{\text{lsp}}(t) = \frac{\mathrm{d}\zeta_{\text{lsp}}}{\mathrm{d}t}$$

c) What is the highest instantaneous velocity that the surface will experience, during the sinusoidal vibration?

The particle velocity, $v(t)$, in the air immediately at the loudspeaker surface, i.e., at $x = 0$, must be identical to the loudspeaker vibration velocity. To be precise, it is the x-component of the particle velocity which equals $v_{\text{lsp}}(t)$. The particle velocity is a vector quantity, but the y-, and z-components are zero for a plane wave (that propagates in the x-direction). We usually simplify the notation and write $v(x,t)$ instead of $v_x(x,t)$ when it is obvious that we have only the x-component.

$$v(x = 0, t) = v_{\text{lsp}}(t). \tag{4}$$

This boundary condition determines the amplitude of the plane wave which is propagating away from the vibrating loudspeaker. (For now, we ignore that there might be a reflected wave returning back from the other end of the tube). If the highest particle velocity is much smaller than the wave speed, $c = 343\,\text{m/s}$, then we have linear conditions for the sound waves.

A plane wave which propagates in the positive x-direction has a sound pressure, p, and particle velocity, v, on these forms:

$$p(x,t) = \hat{p}e^{-jkx}e^{j\omega t} \tag{5}$$
$$v(x,t) = \hat{v}e^{-jkx}e^{j\omega t}, \tag{6}$$

where $\hat{p}$ and $\hat{v}$ are complex constants.
$\hat{p}$ and $\hat{v}$ are related via the acoustic impedance, Z, which for a plane wave in the positive x-direction[1] equals the medium's characteristic impedance:

$$Z = \frac{\hat{p}}{\hat{v}} = \rho_0 c, \tag{7}$$

where ρ_0 is the medium's density (for air, $\rho_0 = 1.2\,\mathrm{kg/m^3}$. The acoustic impedance is in general also a vector quantity.

d) What is the maximum sound pressure in (the harmonic oscillation of) this plane wave?

e) How large is that maximum sound pressure (the "ac-component") compared to the static air pressure (the "dc-component")? In the same way as for the comparison between the particle velocity and the speed of sound; if the maximum sound pressure is much smaller than the static air pressure, then the sound propagation is linear.

f) The time-averaged sound intensity, I, tells how much power, W, in watts, is propagating per m^2 through the tube[2]. Again, the intensity is a vector quantity, but for a plane wave, only one of the three components is non-zero. We can then find the power via these relationships:

$$I = \bar{pv} = \frac{1}{T}\int_0^T p(t)v(t)dt \tag{8}$$
$$W = \int_{S_{\text{total area}}} IdS \rightarrow W_{\text{planewave}} = I_{\text{planewave}} S_{\text{tube}} \tag{9}$$

where *p(t)* and *v(t)* should be the real-valued forms. What is the sound power that propagates through this tube?

[1] For a plane wave which propagates in the negative x-direction, the acoustic impedance is $Z_{negdirection} = \frac{\hat{p}_{negdirection}}{\hat{u}_{negdirection}} = -\rho_0 c$

[2] In acoustics, the symbol W is used for power instead of the more common P in electronics. The reason is that P is easy to confuse with the p for sound pressure.

Figure 2: A dodecahedron loudspeaker from Norsonic AS.

3 Acoustic wave in spherical coordinates

The acoustic wave equation in spherical coordinates has the following solutions for the acoustic quantities $p(r,t)$ and $v(r,t)$ for an outgoing spherical wave:

$$p(r,t) = B\frac{1}{r}e^{j(\omega t - kr)} \tag{10}$$

$$v(r,t) = B\frac{1}{\rho_0 c}\frac{1}{r}\left[1 - \frac{j}{kr}\right]e^{j(\omega t - kr)} \tag{11}$$

B is a constant
$p(r,t)$ is the acoustic pressure [N/m^2]
$v(r,t)$ is the particle velocity in the radial direction [m/s]
ω is the angular velocity [1/s]
$c = 343$ m/s is the propagation speed, "the speed of sound"
$k = \omega/c$ is the wavenumber[1/m]
$\rho_0 = 1{,}21$ kg/m^3 is the density of air

This information is to be used in the following problems:

We approximate a sphere with radius a by a dodecahedron loudspeaker as shown in Figure 2. We assume it has uniform radiation in all directions. The sphere's surface is a membrane which is excited by an unspecified generator providing a radial velocity[3]

$$v(a,t) = v(a)e^{j\omega t}, \tag{12}$$

illustrated schematically in Figure 3.

a) Calculate the pressure and velocity for the spherical wave as a function of r and $v(a)$. (You may remove the time variations.)

b) Define the acoustic impedance Z_a, and calculate it as a function of kr. Discuss the results and provide relevant diagrams.

c) Explain what is meant with acoustic intensity $\overline{I(r)}$, and show that $\overline{I(r)}$ as a function of kr can be written as (the overline means that the intensity value is

[3] Strictly speaking, such a dodecahedron loudspeaker does not have a surface which vibrates uniformly radially, since it has 12 "pistons", that is, 12 circles that vibrate as rigid surfaces. But this is close enough to uniform radial vibration for now.

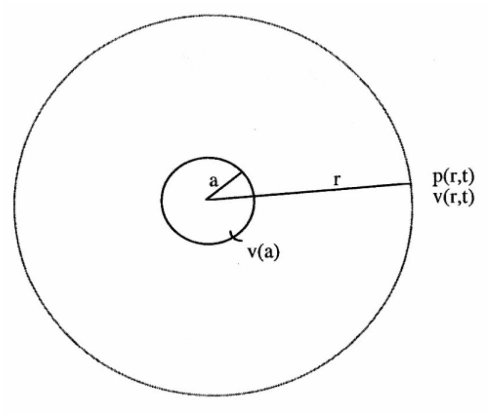

Figure 3: Schematic of the speaker in problem 3.

the time-average of a time-dependent intensity, see further below):

$$\overline{I(r)} = \frac{1}{2}\left(\frac{ka}{kr}\right)^2 |v(a)|^2 \frac{\rho_0 c k^2 a^2}{1 + k^2 a^2}$$

You may need a couple of hints when you embark on solving this subtask:

1. We know that for a purely harmonic wave, the intensity *I(r, t)* is defined as:

$$I(r,t) = \mathrm{Re}\,\{p(r,t)\}\,\mathrm{Re}\,\{v(r,t)\}\,,$$

   where *p(t)* and *v(t)* should be the real-valued forms.
2. You could derive the real parts of p and v from Equations 10 and 11, and then integrate over time. However, it turns out to be quite easier to instead use the fact that the real part of a complex number can be written as $\mathrm{Re}\,\{Z\} = \frac{1}{2}(Z + Z^*)$.
3. If you use the approach suggested above, the multiplication of $\mathrm{Re}\,\{p(r)\}$ and $\mathrm{Re}\,\{v(r)\}$ will give 4 terms, with time dependencies $e^{jwt} * e^{jwt}$, $e^{jwt} * e^{-jwt}$, $e^{-jwt} * e^{-jwt}$, respectively. Please note that when you integrate over time (a whole period of the harmonic time-dependence), the first and third time-dependencies will integrate to 0, whereas the second time-dependence ends up being constant over time.

d) Calculate the total, radiated, power $P_a = W$ and graph it as a function of ka. Discuss the efficiency of the sphere speaker for low and high frequencies.

e) Define the lower limit f_0 as that frequency for which the radiated power is reduced 50%. Find f_0 when $a = 10$ cm.

f) The maximum amplitude s for the oscillating membrane is fixed such that $s/a = 1/100$. Assume the frequency $f = f_0$ and calculate the maximum acoustic power $P_{\max}$ that the sphere can radiate under these conditions.

4 Reflection of plane wave

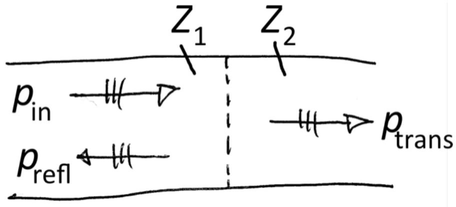

Figure 4: The reflection of a plane wave at a surface

The reflection factor, R, for sound waves is an important characteristic of materials since it tells something about how well the material absorbs sound waves. The reflection factor is defined as

$$R = \frac{\hat{p}_{refl}}{\hat{p}_{in}}$$

and it can in general have a complex, frequency-dependent value, but $|R| \leq 1$. In practice, a more interesting quantity is the absorption coefficient[4], α, defined as

$$\alpha = 1 - |R|^2$$

The value of R for an interface between two media can be derived quite easily for plane waves with perpendicular incidence, and the reflection factor turns out to be

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \tag{13}$$

where Z_1 is the acoustic impedance in medium 1, whereas Z_2 is the acoustic impedance in medium 2, as experienced at the interface. Figure 4 illustrates the situation, with an incident sound wave from the left, denoted p_{in}, and a reflected wave, p_{refl}, and a transmitted wave, p_{trans}, caused by the change in impedance Z at the interface.

The impedance Z_1 is often that of air, $Z_1 = Z_{air} = \rho_0 c = 1.2\,\mathrm{kg/m^3} \cdot 343\,\mathrm{m/s} \approx 412\,\mathrm{kg/(m^2s)}$. It is not so easy to get any intuitive feeling about that value, but the acoustic impedance indicates how strong the reaction force is against a moving surface. Air particles are very light and give a very weak reaction force against moving surfaces. Water, on the other hand, has a much higher value, $Z_{water} \approx 1000\,\mathrm{kg/m^3} \cdot 1500\,\mathrm{m/s} = 1.5 \times 10^6\,\mathrm{kg/(m^2s)}$!

a) What is the reflection factor, R, and absorption coefficient, α, for a plane wave in air which hits a water surface perpendicularly?

b) What is the reflection factor, R, and absorption coefficient, α, for a plane wave in water which hits the air surface perpendicularly? What does a negative sign of the reflection factor mean?

We can realize, from Eq. (13), that sound absorbing materials need to have impedances, Z_2, which are as close as possible to that of air. That implies that sound absorbing materials must be very porous and/or soft. Glassfibre wool and mineral wool, which are used for heat insulation of houses, are very efficient sound absorbing materials.

[4] The terms "factor" and "coefficient" are used somewhat unsystematically for reflection, transmission and absorption, even though they have been defined very clearly in ISO standards: a "factor" can not have a unit, whereas a "coefficient" might have a unit, such as "temperature coefficient" for a resistor which might have the unit $\Omega/^{\circ}C$. So, α, as we use it here, should be called "absorption factor", which is the case in the Norwegian standard, but internationally, the term "absorption coefficient" has stuck.

Such fibrous materials need to be covered by some surface material, and we can study one simple construction: a free-hanging thin plastic foil in front of a mineral wool, and we assume that the mineral wool acts as a perfect absorber. This is illustrated in Fig. 5.

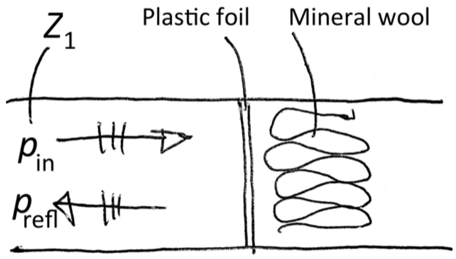

Figure 5: The reflection of a plane wave by a sound absorbing material which is covered by a plastic foil

c) We can assume that for the case in Fig. 5, $Z_2 = \rho_0 c + j\omega m''$ where m'' is the mass per surface area of the plastic foil. This particular foil has $m'' = 20\,\mathrm{g/m^2}$ (which could be a $20\,\mu\mathrm{m}$ thick plastic foil). Calculate and plot the magnitude of the reflection factor and the absorption coefficient as a function of frequency. Choose, e.g., frequencies that are logarithmically distributed between $100\,\mathrm{Hz}$ and $10\,\mathrm{kHz}$.

This particular absorber has a very high absorption coefficient for low frequencies. Is its high-frequency performance affected by the plastic foil?